

Information Retrieval Fusion Method Based on Cooperative Games

final project advanced IR course

<https://github.com/kfir7755/advanced-IR-project>

Kfir Eliyahu

kfir.eliyahu@campus.technion.ac.il

Guy Shoef

guy.shoef@campus.technion.ac.il

Abstract

In information retrieval, various rankers aim to order documents from a large corpus based on their relevance to a given query. Fusion methods combine ranked lists from multiple rankers to produce a single, more effective ranking. In this work, we present two novel variants that challenge the focus of existing research on designing new scoring functions. Instead, we leverage cooperative game theory and Shapley Values to determine the optimal weighting for each ranker. Our approach demonstrates improved empirical performance over prior methods, supported by statistical evidence across multiple tracks of the TREC-03 datasets.

1 Introduction

In **Information Retrieval (IR)**, the fundamental challenge is to identify the most relevant documents for a given query from a large collection. Various ranking models have been developed to address this problem, including vector space models (e.g., TF-IDF ranker [1]), probabilistic ranking functions (e.g., BM25 [2]), relevance-based language models (e.g., RM3 [3, 4]), and deep learning based rankers (e.g., RankGPT [5]). Each ranking approach has its strengths and limitations, and a common strategy for improving retrieval effectiveness is to aggregate multiple rankers into a single, more robust ranking function. This process, known as **fusion**, combines the scores from different rankers to produce a unified ranking that better reflects document relevance to the query. Most fusion methods rely on a scoring function to normalize score distributions across different rankers. These functions can be categorized into two types:

1. **Rank-based scoring**, which considers only the relative ordering of documents without using their raw scores.
2. **Score normalization**, which adjusts ranker scores to a common scale.

A key limitation of existing fusion models is that they typically assign uniform weights to all rankers, despite potential differences in their effectiveness. While some approaches introduce weighting mechanisms, most prior work has focused on exploring alternative scoring functions rather than optimizing ranker weighting. Another

limitation is that when weights are assigned, they are often derived solely from the ranker’s performance on a training dataset, typically measured using metrics such as Mean Average Precision (MAP) or Precision (P) [6].

In this work, we propose a novel weighting approach for ranker fusion based on **cooperative game theory**, specifically leveraging a simplified variant of **Shapley values** [7]. Unlike prior methods that determine weights based on a ranker’s standalone performance, our approach assigns weights based on each ranker’s contribution to the fusion model as a whole. We demonstrate that our weighting method provides statistically significant performance improvements across three different TREC datasets.

2 Related Work and Preliminaries

There are two topics related to our work: Fusion Methods in IR and Cooperative Game Theory & Shapley Value.

2.1 Fusion Methods in IR

Fusion in information retrieval relies on having n rankers. Each ranker i produces a document list L_i along with a corresponding score function $s(L_i)$, which assigns a score to each document based on its relevance. Typically, each list L_i contains only the top- k documents (usually $k = 1000$). The rank of a document d in the i th ranker is denoted by $r_{L_i}(d)$. The fusion scoring rule is denoted by $F_{\text{fusion-method}}$.

There are numerous methods and frameworks for fusing document lists retrieved from the same corpus in response to a query. Two of the most common methods are CombSUM and CombMNZ [6]. CombSUM is defined as the average score output per ranker:

$$F_{\text{CombSUM}}(d) \triangleq \frac{1}{n} \sum_{i=1}^n s_{L_i}(d) \quad (1)$$

CombMNZ is a nonlinear method that multiplies the sum of scores by the number of rankers that include document d in their list:

$$F_{\text{CombMNZ}}(d) \triangleq |\{L_i : d \in L_i\}| \sum_{i=1}^n s_{L_i}(d) \quad (2)$$

CombSUM also has a weighted version, called **WeightedCombSUM**, in which the uniform averaging is replaced by ranker-specific weights:

$$F_{\text{WeightedCombSUM}}(d) \triangleq \sum_{i=1}^n w_i s_{L_i}(d) \quad (3)$$

Regarding scoring functions, the two main rank-based variants are **Borda** [8] and **RRF** [9]. Borda assigns a score based on the document’s rank linearly:

$$s_{L_i}(d) \triangleq k - r_{L_i}(d) \quad (4)$$

RRF (Reciprocal Rank Fusion) is defined as:

$$s_{L_i}(d) \triangleq \frac{1}{v + r_{L_i}(d)} \quad (5)$$

where v is a free parameter, usually set to $v = 60$.

For non-rank-based scoring functions, normalization of each ranker’s scores is typically applied. The main normalization methods include **MinMaxNorm**, **SumNorm**, and **ZNorm** [6]. These methods adjust the scores in each list according to a specific normalization strategy. Sometimes the normalization is applied only to the top portion of the list (e.g., using the sum of the top 100 scores rather than all 1000).

2.2 Cooperative Game Theory and Shapley Value

In cooperative game theory, multiple players collaborate to achieve a common goal, and the total reward is distributed among the players (i.e., coalition members). A central problem is to determine a fair distribution of rewards that satisfies properties such as efficiency (all available reward is distributed) and symmetry (the order of players does not affect the distribution). A cooperative game with n players is defined by a player set $N = \{1, 2, \dots, n\}$ and a characteristic value function $v : 2^N \rightarrow \mathbb{R}$, which assigns a total reward to each possible coalition, with $v(\emptyset) = 0$.

The **Shapley Value** [7], introduced by Lloyd Shapley (a 2012 Nobel Prize winner in Economics), is one of the most well-known methods for fairly distributing rewards. The Shapley Value is the unique distribution rule that satisfies the properties of efficiency, symmetry, the dummy player (a player that contributes nothing receives nothing), and additivity. It assumes all the players collaborate and the value it gives to the i ’th player is the average contribution to N for the total permutations N has. Formally, it can be described as the following:

$$\varphi_i(v) = \frac{1}{n!} \sum_R [v(P_i^R \cup \{i\}) - v(P_i^R)] \quad (6)$$

where the sum is taken over all permutations R of N and P_i^R denotes the set of players that precede player i in permutation R [10].

This approach has been game-changing for feature importance and AI explainability. However, computing the exact Shapley Values is NP-hard [11], which has led to various approximation methods, such as SHAP [12]. Specific approximations exist for particular models, such as TreeSHAP [13] for decision trees and Deep SHAP [12] for neural networks.

3 Method

We propose two novel variants for weighting in the fusion framework of WeightedCombSUM. Both approaches reduce the NP-hard computation of Shapley Values by summing only over the permutations in which the i -th ranker appears second. In this case, the formulation for the approximated Shapley Values is:

$$\begin{aligned} \tilde{\varphi}_i(v) &= \frac{1}{n} \sum_R [v(P_i^R \cup \{i\}) - v(P_i^R)] \\ &= \frac{1}{n} \sum_{j \neq i} [v(\{j, i\}) - v(\{j\})] \end{aligned} \quad (7)$$

Our two variants differ in how they define the characteristic function v and the computation of Shapley Values. Our baseline variant, **diffscore**, treats v as the performance metric (either MAP or P) of the ranker when used individually. When used in a fusion model, it follows previous work by weighting each ranker based on its standalone performance.

Our second variant, **ReLUdiffscore**, assumes that v represents the optimal performance achievable by combining the given rankers using WeightedCombSUM. Since the exact value of v is infeasible to compute, we approximate it using prior weighting methods. If the true v were used, Shapley Values would always be non-negative. However, with our approximation, negative values are possible. To better approximate the real Shapley Values, we apply the ReLU function (i.e., $\max(x, 0)$) to the diffscore weights, ensuring non-negative contributions and improving the reliability of the weighting scheme.

4 Dataset

In our study, we utilized existing runs from rankers submitted to **TREC-03** [14, 15] across three different tracks: **Robust**, **Ad hoc**, and **Routing**. Each track consists of a document collection of approximately 500,000 documents. For the ROBUST track, 30 rankers were submitted, while the Ad hoc and Routing tracks had 40 and 34 rankers, respectively. Additionally, the Robust track contains 100 queries, while the Ad hoc and Routing tracks each contain 50 queries.

5 Experimental Setup

We conducted a 5-fold cross-validation on each of the TREC-03 tracks (Robust, Ad hoc and Routing).

- The metrics for weighting used in prior work are MAP@100 and P@10.
- The fusion scoring functions tested are Borda, RRF, MinMaxNorm, and SumNorm.
- The weighting methods compared are ReLUdiffscore, diffscore, metric-based (i.e., prior work), and uniform weighting (CombSUM).

For each fold, we evaluated performance on the test set using MAP@100 and P@10, then averaged the results across all five folds.

6 Results

The results of the experiments can be viewed in 1, 2, 3 for TREC-03 Robust, TREC-03 Ad hoc, TREC-03 Routing, respectively. Statistical significance for all the results was determined with a two-tailed Student's t-test at $\alpha = 0.05$. A positive significant difference relative to the metric weight method is indicated by a † and relative to the uniform weight method by a ‡ . Conversely, a negative significant difference is marked with a ‡ when compared to the metric weight method and with a † when compared to the uniform weight method. In **bold** the best overall metric among the weighting methods. MAP refers to MAP@100 and P refers to P@10.

Table 1: The table shows the average results over 100 test queries from the **TREC-03 Robust** Track qrels using 5-fold cross-validation.

metric used	weight method evaluation metric fusion method	ReLUdiffscore		diffscore		metric		uniform	
		MAP	P	MAP	P	MAP	P	MAP	P
MAP	Borda	0.243	0.445	0.230	0.424	0.238	0.438	0.236	0.439
	MinMaxNorm	0.252 ‡	0.452	0.252 ‡	0.451	0.244	0.445	0.243	0.445
	RRF	0.252 ‡	0.450	0.252 ‡	0.450	0.246	0.446	0.244	0.446
	SumNorm	0.259 ‡	0.455	0.035 ‡	0.094 ‡	0.249	0.457	0.248	0.452
P	Borda	0.239	0.440	0.239	0.440	0.237	0.441	0.236	0.439
	MinMaxNorm	0.248 ‡	0.455 ‡	0.248 ‡	0.455 ‡	0.243	0.444	0.243	0.445
	RRF	0.250 ‡	0.455 ‡	0.250 ‡	0.455 ‡	0.245	0.445	0.244	0.446
	SumNorm	0.258 ‡	0.471	0.030 ‡	0.069 ‡	0.248	0.454	0.248	0.452
overall		0.250	0.453	0.192	0.355	0.244	0.446	0.243	0.446

Table 2: The table shows the average results over 50 test queries from the **TREC-03 Ad hoc** qrels using 5-fold cross-validation.

metric used	weight method evaluation metric fusion method	ReLUdiffscore		diffscore		metric		uniform	
		MAP	P	MAP	P	MAP	P	MAP	P
MAP	Borda	0.242 ‡	0.758 ‡	0.243 ‡	0.762 ‡	0.237	0.748	0.233	0.734
	MinMaxNorm	0.250 ‡	0.760 ‡	0.250 ‡	0.760 ‡	0.242	0.744	0.236	0.744
	RRF	0.248 ‡	0.756 ‡	0.248 ‡	0.756 ‡	0.238	0.748	0.232	0.732
	SumNorm	0.245 ‡	0.744	0.199 ‡	0.668 ‡	0.228	0.734	0.215	0.718
P	Borda	0.240 ‡	0.754 ‡	0.241 ‡	0.758 ‡	0.236	0.744	0.233	0.734
	MinMaxNorm	0.248 ‡	0.758	0.248 ‡	0.758	0.240	0.748	0.236	0.744
	RRF	0.243 ‡	0.750 ‡	0.243 ‡	0.750 ‡	0.236	0.748	0.232	0.732
	SumNorm	0.244 ‡	0.748	0.209 ‡	0.684	0.225	0.730	0.215	0.718
overall		0.245	0.754	0.235	0.737	0.235	0.743	0.229	0.732

Table 3: The table shows the average results over 50 test queries from the **TREC-03 Routing** qrels using 5-fold cross-validation.

metric used	weight method evaluation metric fusion method	ReLUdiffscore		diffscore		metric		uniform	
		MAP	P	MAP	P	MAP	P	MAP	P
MAP	Borda	0.254 [†]	0.698	0.254 [†]	0.696	0.250	0.692	0.244	0.676
	MinMaxNorm	0.249 ^{††}	0.692	0.249 ^{††}	0.692	0.243	0.682	0.241	0.684
	RRF	0.252	0.690	0.252	0.690	0.248	0.684	0.247	0.684
	SumNorm	0.251	0.696	0.251	0.694	0.247	0.682	0.245	0.680
P	Borda	0.251	0.694	0.249	0.696	0.249	0.688	0.244	0.676
	MinMaxNorm	0.250 ^{††}	0.694	0.250 ^{††}	0.694	0.243	0.686	0.241	0.684
	RRF	0.251	0.682	0.251	0.682	0.248	0.684	0.247	0.684
	SumNorm	0.248	0.684	0.224 ^{††}	0.660	0.247	0.684	0.245	0.680
overall		0.251	0.691	0.247	0.688	0.247	0.685	0.244	0.681

7 Conclusions

Our results demonstrate that ReLUdiffscore is highly effective, achieving the best performance across all tracks. It not only outperforms previous methods empirically, but also shows strong statistical evidence of improvement on the Robust and Ad hoc tracks. There were less statistical significances on the Routing track, despite ReLUdiffscore still achieving the best empirical performance.

ReLUdiffscore performs particularly well when paired with the MinMaxNorm scoring function, showing statistically significant improvements in MAP over both Uniform and Metric weighting methods across all tracks.

In contrast, while diffscore often achieves results similar to ReLUdiffscore, it has major failures in certain cases, such as with the SumNorm scoring function. Analysis of the weights assigned by diffscore in those cases reveals that it frequently assigns negative weights to rankers, potentially excluding documents that appear in multiple ranked lists from the final fused list.

Overall, ReLUdiffscore appears to be the superior variant. This supports our hypothesis that approximating Shapley Values by "grounding" diffscore weights — ensuring they maintain properties similar to the true Shapley Values, in our case- non-negativity, leads to a more robust fusion method.

8 Future Work

There is significant potential for further research, particularly in refining approximations to the true Shapley Values.

One possible improvement to the ReLUdiffscore variant is modifying its "grounding" mechanism. Instead of applying ReLU to the final weight, we could apply ReLU to each term in the summation, leading to the following formulation:

$$\tilde{\varphi}_i(v) = \frac{1}{n} \sum_R \text{ReLU} [v(P_i^R \cup \{i\}) - v(P_i^R)] \quad (8)$$

Additionally, improving the approximation of Shapley Values could involve exploring better sampling tech-

niques for permutations of N . Our current method restricts permutations to a very specific subset—where the i -th ranker is always second. However, there may exist more effective strategies for sampling permutations that yield better approximations to the true Shapley Values in the context of ranker weighting.

References

- [1] G. Salton and C. Buckley, “Term-weighting approaches in automatic text retrieval,” *Information processing & management*, vol. 24, no. 5, pp. 513–523, 1988.
- [2] S. E. Robertson, S. Walker, S. Jones, M. M. Hancock-Beaulieu, M. Gatford, *et al.*, “Okapi at trec-3,” *Nist Special Publication Sp*, vol. 109, p. 109, 1995.
- [3] V. Lavrenko and W. B. Croft, “Relevance-based language models,” in *ACM SIGIR Forum*, vol. 51, pp. 260–267, ACM New York, NY, USA, 2017.
- [4] N. Abdul-Jaleel, J. Allan, W. B. Croft, F. Diaz, L. Larkey, X. Li, M. D. Smucker, and C. Wade, “Umass at trec 2004: Novelty and hard,” in *Proceedings of TREC-13*, pp. 715–725, 2004.
- [5] W. Sun, L. Yan, X. Ma, S. Wang, P. Ren, Z. Chen, D. Yin, and Z. Ren, “Is chatgpt good at search? investigating large language models as re-ranking agents,” *arXiv preprint arXiv:2304.09542*, 2023.
- [6] Y. Anava, A. Shtok, O. Kurland, and E. Rabinovich, “A probabilistic fusion framework,” in *Proceedings of the 25th ACM International on Conference on Information and Knowledge Management*, pp. 1463–1472, 2016.
- [7] L. S. Shapley *et al.*, “A value for n-person games,” 1953.
- [8] J. A. Aslam and M. Montague, “Models for metasearch,” in *Proceedings of the 24th annual international ACM SIGIR conference on Research and development in information retrieval*, pp. 276–284, 2001.
- [9] G. V. Cormack, C. L. Clarke, and S. Buettcher, “Reciprocal rank fusion outperforms condorcet and individual rank learning methods,” in *Proceedings of the 32nd international ACM SIGIR conference on Research and development in information retrieval*, pp. 758–759, 2009.
- [10] C. Levinger, N. Hazon, and A. Azaria, “Efficient computation and estimation of the shapley value for traveling salesman games,” *IEEE Access*, vol. 9, pp. 129119–129129, 2021.
- [11] T. P. Michalak, K. V. Aadithya, P. L. Szczepanski, B. Ravindran, and N. R. Jennings, “Efficient computation of the shapley value for game-theoretic network centrality,” *Journal of Artificial Intelligence Research*, vol. 46, pp. 607–650, 2013.
- [12] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” *Advances in neural information processing systems*, vol. 30, 2017.
- [13] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee, “From local explanations to global understanding with explainable ai for trees,” *Nature machine intelligence*, vol. 2, no. 1, pp. 56–67, 2020.
- [14] E. M. Voorhees and D. Harman, “Overview of trec 2003,” in *Trec*, pp. 1–13, 2003.
- [15] E. M. Voorhees *et al.*, “Overview of the trec 2003 robust retrieval track,” in *Trec*, pp. 69–77, 2003.